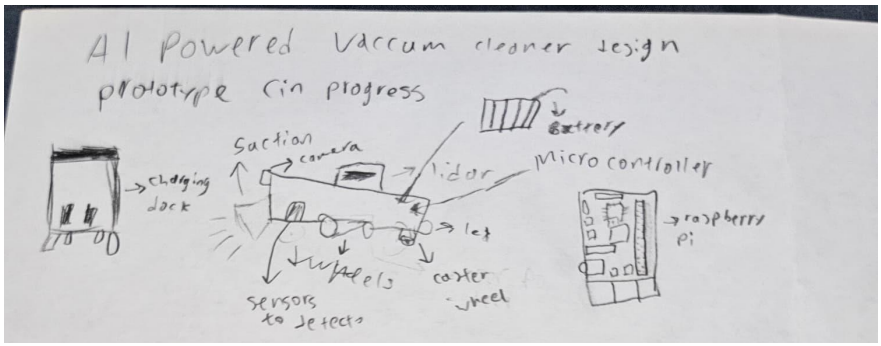
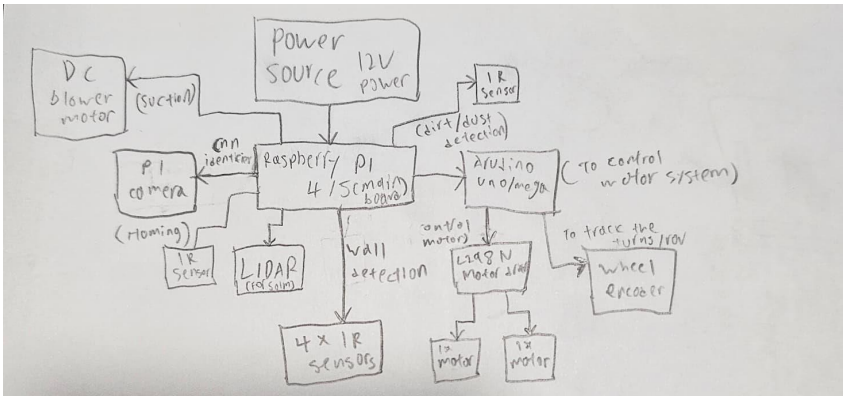
University of Hertfordshire UH

Logbook 1

Name:	**Jovian Lim Zheng Yuan**
Week Date:	**Part A -Meeting 1 of 8**

(*The above information is for you to keep track of which week this log is applicable for. This information is not required in the online logbook as the system will find your name, your supervisor's name, and take the submission date, automatically. You must submit the online log in the relevant week. Logs* ***cannot*** *be backdated to missed weeks.)*

Meeting date: *(please note if no meeting is planned this week)*	**Date and day** **28/05/26 , Thursday**
Work Undertaken this week: *(No more than 4000 characters, including spaces)*	**From the project title that you have selected, do a research on similar projects (at least 4) and summarize your finding on each one.** • **Roomba Max 705 (USA):** Roomba uses the LiDAR mapping and a camera to build accurate floor plans and avoid low-lying obstacles in dim lighting. With that its able to work in the dark as well , and cleans when no light is around. • **Roborock (2025 arm) (china):** Roborock has an additional robotic arm to physically remove small items blocking its path when cleaning. Such as socks, tissue and etc. • **DEEBOT X11 (china):** High-power suction with software that learns user preferences and customizes cleaning tasks. It suction pressure is 19 500pa compare to other robot vacuum cleaner that is low suction power. And learns on the go after every complete cleaning schedule. • **Samsung Jet Bot AI+ Steam Ultra (South Korea):** Samsung Jet Bot have a combines LiDAR and 3D vision for robust object detection and with it unique design it can traverse or roll over minor obstacles hence it wont affect the cleaning operations/ tasks. Summary for the findings: All four products rely on LiDAR as the core navigation sensor, which confirms it as the right choice for this project. Each product takes a different approach to handling obstacles and floor types — from software avoidance, to physical manipulation, to rolling over them entirely. The shared theme across all four is using AI to make decisions on the go rather than following a fixed cleaning path. This project adopts the same principle, using a custom SLAM system and a CNN classifier to make real-time decisions based on the environment.

Problems encountered: *(No more than 4000 characters, including spaces)*	**What is the information that are not clear or sufficient for your intended project. If no, you can skip this.** **Due to not using ROS2 for SLAM I have to code using a different software as ros2 has many bugs to interpret. Plus TurtleBot wont be suitable choice as its too compact.** **Other hardware components like Arduino uno have to check with the lab technician if they have spare.**
Tasks agreed for upcoming week: *(No more than 4000 characters, including spaces)*	**What are your plan actions/task until the next progress meeting.** **I will have a clearer mind on how the system works and create the block diagram with all the components needed for my project plus a follow up on my sketch on how my vacuum cleaner like look like. To add on with the major impact to create slam without ROS, I will research the alternatives on it**
Figures: *(Upload image files in the online Log Form, but feel free to keep notes here about which images you intend to upload. Use "Figure Caption" information as the filenames, e.g., "Figure 1 – A screenshot of the initial constrained Sketch in CATIA.jpg")*	**1) Provide / Sketch an artistic view of your project. A rough one will do. You can use any software like PPT to design this sketch**  *Figure 1: Sketch of Project* **2) Update Block diagram of your system**  *Figure 2: Block Diagram* **3) Key software codes/modules that need to be developed** **There will be multiple codes to integrate into this project like such as to call out the hardware, navigation, ai logical thinking and for app communication plus some are utilities code for the backend. Below I**

	have split them to groups • **Hardware** - lidar.py its to call the lidar out to get the distance and views - motor.py control the motor speed and direction -camera.py calls the camerause for cnn -beacon.py for homing when it needs to recharge or go home - Sensors.py reads the sensors, battery voltage, obstacle avoidance -odmetry.py mainly for reading the encoder cycles and show the axis of the robot • **Navigation** -Slam.py updated the map from the lidar scanning plus odmetry -path-planner.py it generates the path to clean from the map -Obstacle Avoidance.py for wall avoidance -controller.py to track the robot between point to point • **AI aspects** -surface_classifier.py identify the tiles on the floor appearance like carpet or wood or others -dirt_detector.py sees dirt/dust on maps and marks as dirt section -suction_control.py controls the power of the motor based on the floor type -train_classifier.py trains the model • **App codes** - Server.py for map logging - Mqtt_cilent.py for map updates - Map-encoders to convert numpy to json file for display - Session_logger.py save all logs and history of clean history • **Utilities** - Visualizer.py display the map on the app - Logger.py in the terminal - Simulator.py testing the programme to see hows the lidar work with - Arduino_flasher.py communicate with raspberry **4) Any Quality and Regulatory issued to be addressed?** ● **Quality:** - For hardware the vacuum cleaner may overheat if is overcharge and motor safety - Electrical safety due to using lipo battery it needs be well protected and prevent it from short circuit and fire risk ● **Regulatory: currently there no regulatory issues but will check in the future but accessing the camera will breach personal space since it will capture the house during mapping.**